Jianna Shao

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EDUCATION

Wesleyan University, Middletown, CT — Expected May 2029
B.A. in Computer Science, Art Studio, and Integrated Design and Engineering
Relevant Coursework: Multivariable Calculus, General Physics I, Working with MATLAB, Mechanical Design and Engineering, Product Design I, How to Design Programs, Discrete Mathematics (in progress)

RESEARCH EXPERIENCE

Undergraduate Student Researcher, Wesleyan University, Middletown, CT — June 2026 – Present
Wire Arc Additive Manufacturing (WAAM) | Advisor: Elizabeth Chang-Davidson

- Lead a study on how curved toolpaths and process parameters (wire feed rate, travel speed, toolpath radius) affect steel bead geometry in wire arc additive manufacturing (WAAM), as
- Program an ABB robot with a Fronius GMAW welder in RobotStudio to deposit steel beads along curved toolpaths.
- Collected and analyzed 1,000+ bead width measurements in MATLAB across a matrix of process parameters.
- Section samples with a waterjet cutter, mount them in acrylic, and grind, polish, and etch them for optical microscopy.
- Examine cross-sectional microstructure to validate caliper measurements and identify porosity defects.
- Developed data-driven normalization methods and non-linear regression models that predict bead width with ~90% accuracy and can be refit for other processes and materials.
- Presented findings in two poster sessions and two oral presentations; first-authoring a manuscript currently under advisor review.

PUBLICATIONS AND PRESENTATIONS

Shao, J., Blacker, A., & Chang-Davidson, E. Predictive Modeling of Bead Geometry at Curvatures in Wire Arc Additive Manufacturing. *Manuscript in preparation.*
Shao, J. "Pointless: The Curves and the Beads Talk." Two poster presentations and two oral presentations, Wesleyan University, Middletown, CT, 2026.

HONORS AND AWARDS

Spark Prize for research communication, Wesleyan Summer Research Program — 2026
Wesleyan Summer Research Fellow — 2026

TECHNICAL SKILLS

Laboratory: WAAM (Fronius GMAW), ABB robot programming (RobotStudio), waterjet cutting, metallographic sample preparation (mounting, grinding, polishing, etching), optical microscopy; chemical handling training
Computational: MATLAB (data analysis, normalization, regression modeling), GitHub, Godot
Design & CAD: Rhino 8, Onshape, Blender, Figma, Adobe Illustrator
Languages: English (native), Mandarin (advanced), French (beginner)

ADDITIONAL EXPERIENCE

Graphic Design and Marketing Assistant, Olin Library, Wesleyan University — Sept 2026 – Present

- Work directly with library departments to translate their goals into print and digital materials, managing multiple requests on deadline.

Lead Prop Designer, Spike Tape (student theater), Wesleyan University — Sept 2026 – Present

- Direct the design, sourcing, and fabrication of stage props within a $150 budget; supervise an assistant.

Media Assistant, Valley101, Las Vegas, NV — January 2026

- Covered 7+ keynotes and panels at CES 2026, documenting emerging technology trends.

E-commerce and Marketing Intern, Shanghai Fire Beast Technologies, Shanghai — April – June 2025

- Conducted market research on 10+ potential products, defining target customers and analyzing competitors.